

Early Stopping and Batch Normalization

Deep Learning

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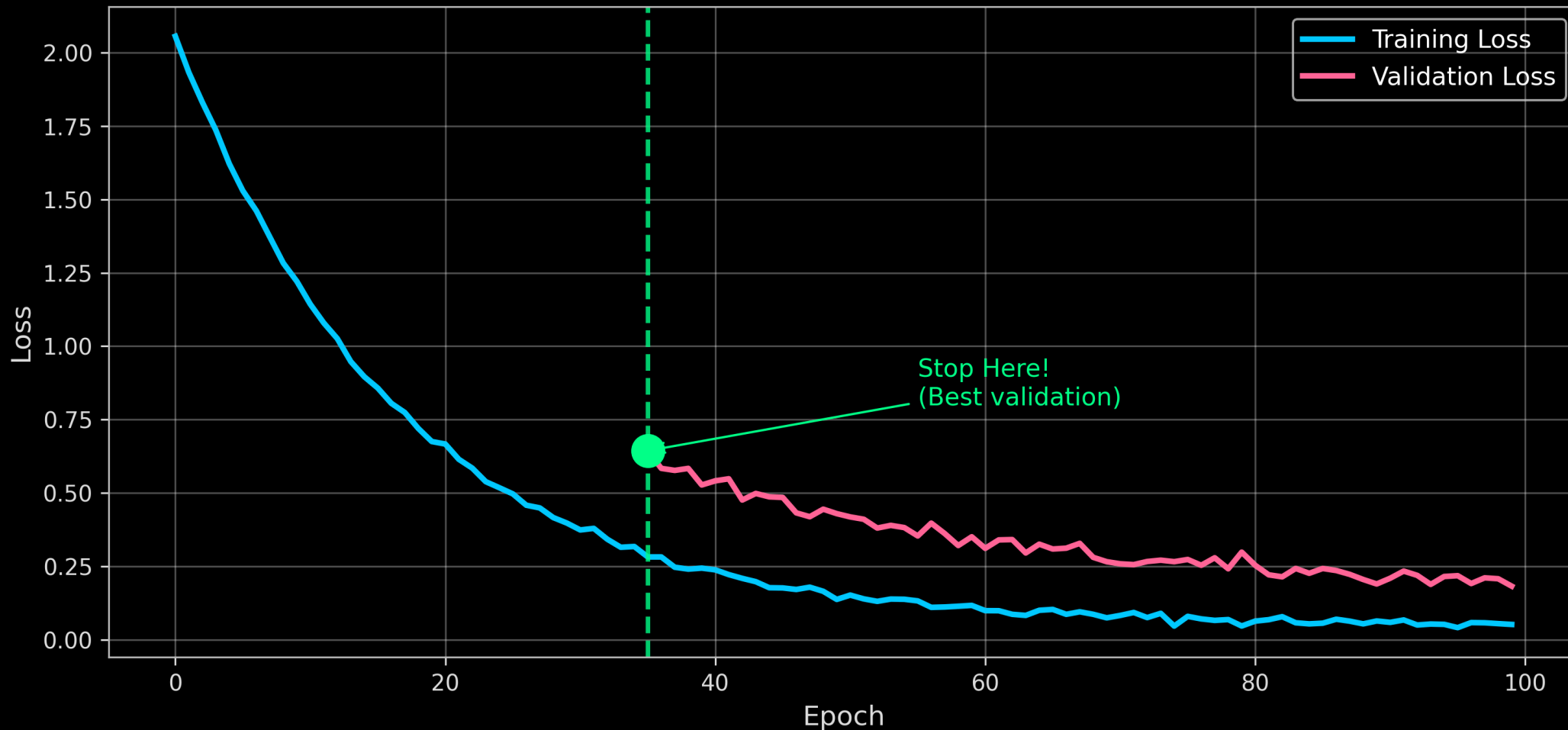
Contents of This Video

In this video, we will cover:

- Early stopping: when to halt training
- Implementing early stopping with patience
- Batch normalization: what and why
- How batch norm stabilizes training
- Practical usage of both techniques

Early Stopping: The Idea

Early Stopping: Stop When Validation Loss Stops Improving



Implementing Early Stopping

Patience: How many epochs to wait before stopping

Algorithm:

1. Track best validation loss seen so far
2. If validation loss improves, reset patience counter
3. If no improvement for `patience` epochs, stop
4. Restore best model weights

Key parameters:

- **Monitor:** What metric to track (usually validation loss)
- **Patience:** How many epochs to wait (typically 5-10)
- **Restore best weights:** Return to the model that had the lowest validation loss

Batch Normalization: The Problem

Deep networks have unstable training:

- Each layer's inputs depend on previous layers
- As weights update, input distributions shift
- Layers must constantly adapt to new distributions
- Called “internal covariate shift”

Symptoms:

- Need very small learning rates
- Careful initialization required
- Training can be slow or unstable

Batch Normalization: The Solution

Normalize activations within each mini-batch:

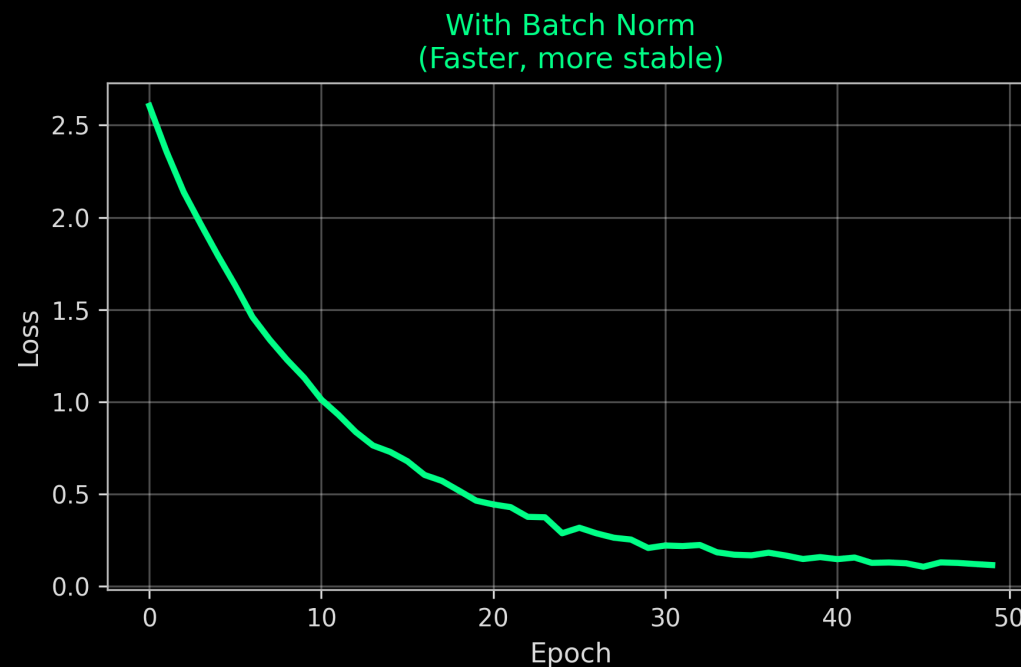
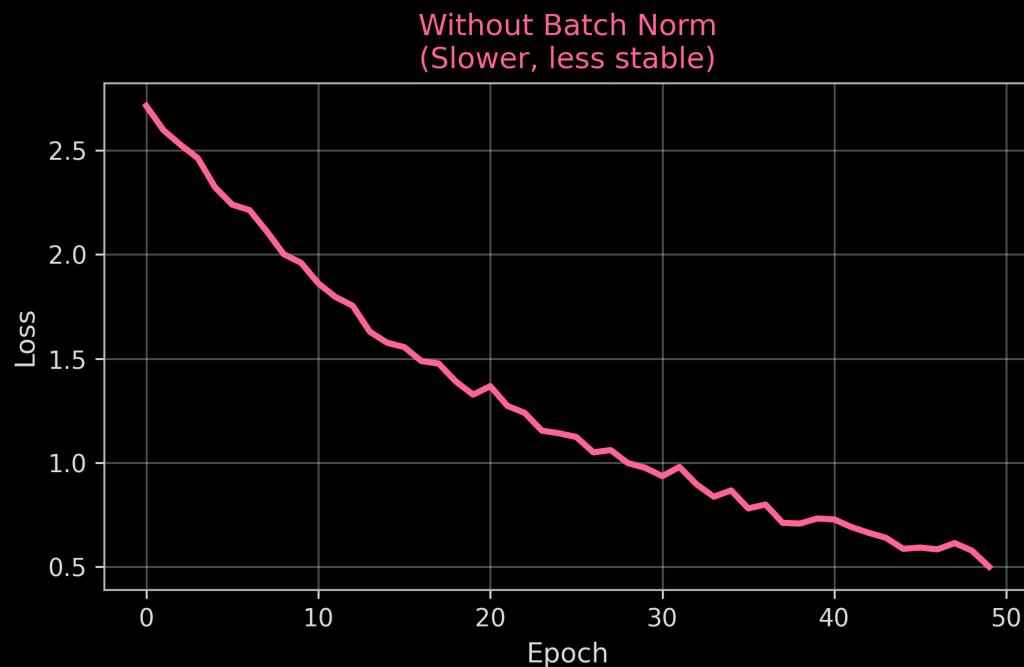
$$\hat{x} = \frac{x - \mu_{batch}}{\sqrt{\sigma_{batch}^2 + \epsilon}}$$

$$y = \gamma \hat{x} + \beta$$

Where:

- $\mu_{batch}, \sigma_{batch}^2$ = mean and variance of batch
- γ, β = learnable scale and shift parameters
- ϵ = small constant for numerical stability

How Batch Norm Helps



Benefits:

- Faster convergence (can use higher learning rates)
- More stable training (less sensitive to initialization)
- Mild regularization effect

Using Batch Normalization

Typical placement in the network:

Order	Component
1	Linear layer (fully connected or convolutional)
2	Batch normalization
3	Activation function (ReLU, etc.)

Position: After linear layer, before activation (debated, both work)

Note: Batch norm has different behavior during training vs. inference

Combining Techniques

A well-regularized training setup:

Layer/Technique	Purpose
Linear layer (256 neurons)	Feature transformation
Batch normalization	Stabilize activations
ReLU activation	Non-linearity
Dropout (rate 0.3)	Regularization
Linear layer (128 neurons)	Feature transformation
Batch normalization	Stabilize activations
ReLU activation	Non-linearity
Dropout (rate 0.3)	Regularization
Output layer (softmax)	Classification

What We've Covered

In this video, we've learned:

- Early stopping: halt training when validation loss stops improving
- Use patience parameter to avoid premature stopping
- Batch norm: normalize activations for stable training
- Batch norm enables higher learning rates and faster convergence
- Combine batch norm + dropout + early stopping for robust training